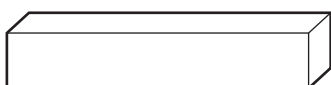


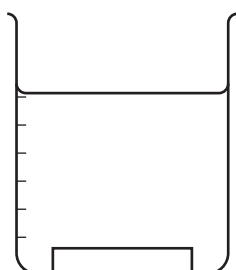
- 6 A student carried out an investigation to estimate the water potential of potato tissue. The main steps in the procedure and in the analysis of results are outlined in Fig. 6.1.

beaker	concentration of sucrose solution / mol dm ⁻³
1	0.0
2	0.1
3	0.2
4	0.3
5	0.4
6	0.5

Six different concentrations of sucrose solution were prepared and an equal volume of each was placed in a labelled beaker.



Six equal-sized blocks of potato tissue were cut out of the same potato, blotted dry and weighed.



One potato block was immersed in the solution in each beaker for 30 minutes.

After this time, the block was removed, blotted dry and reweighed.



The experiment was repeated twice.



The mean percentage change in mass of potato tissue was calculated for each concentration of sucrose used.



A graph was drawn of mean percentage change in the mass of potato tissue against concentration of sucrose.

Fig. 6.1

- (a)** Explain why the different concentrations of sucrose result in different mean percentage changes in mass of potato tissue.

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- (b)** State how the graph is used to estimate the water potential of the potato tissue.

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[Total: 4]